

Business Insights & Recommendations Report

~ End-to-End Data Pipeline & Predictive Analytics - Stock Market

1. Project Overview

This project digs into six years of daily trading data for four big tech stocks—Apple, Microsoft, Google, and Amazon. I pulled all the numbers straight from Yahoo Finance using the `yfinance` library. The idea here is to break down past market trends, create useful technical indicators, and build a machine learning model that predicts what the stock will do the next day. On top of that, the project comes with interactive dashboards to help with business analysis and smarter investing decisions.

2. Data Collection & Cleaning

- **Source:** Yahoo Finance API via `'yfinance'`
- **Stocks Analyzed:** Apple (AAPL), Microsoft (MSFT), Alphabet (GOOGL), Amazon (AMZN)
- **Time Period:** January 2019 - Present
- **Rows collected:** Approximately 7,000+ daily trading records
- **Cleaning performed:**
 - Removed duplicate records
 - Converted date and numerical columns to appropriate data types
 - Sorted data chronologically for each stock
 - Handled missing values created during feature engineering
 - Removed incomplete rows after rolling calculations
- **Result:** A clean, consistent dataset ready for exploratory analysis and machine learning.

3. Key EDA Findings

- **Microsoft (MSFT)** maintained the highest average stock price during the analysis period.
- **Apple (AAPL)** recorded the highest average trading volume, indicating greater market liquidity.
- Daily stock returns were centered around zero with occasional extreme values, suggesting that short-term market movements are highly volatile and difficult to predict.
- The correlation analysis showed that moving averages and lagged closing prices were highly correlated with the current stock price.
- Rolling volatility successfully identified periods of increased market uncertainty and risk across all stocks.
- Overall, all four technology stocks demonstrated strong long-term growth despite temporary market corrections.

4. Feature Engineering

- Engineered technical indicators used as model inputs:
 - Daily Return - Percentage daily price movement
 - Moving Average (7-Day) - Captures short-term price trends
 - Moving Average (30-Day) - Captures long-term price trends
 - Rolling Volatility (7-Day) - Measures short-term market risk
 - Relative Strength Index (RSI-14) - Momentum indicator used to identify overbought and oversold conditions
 - Lag Features (Close t-1, t-2, t-3) - Capture temporal dependencies between previous trading sessions

These engineered features improve the model's ability to identify meaningful market patterns beyond raw price data.

5. Predictive Model & Evaluation

Three regression models were trained using a **chronological train-test** split to simulate real-world forecasting conditions.

MODEL	RMSE	MAE	R ²
Linear Regression	2.019	1.426	-0.0595
Random Forest	1.986	1.387	-0.0242
XGBoost	2.091	1.474	-0.1364

Selected model: Random forest as lowest RMSE on held-out test data.

Why the simple model won: All the models ended up with negative R² values, but Random Forest still came out on top—it had the lowest prediction error (both RMSE and MAE). So, for short-term stock return prediction, it performed best. Still, these results show just how hard it is to predict daily stock market moves. There are too many wild cards: economic news, investor moods, sudden headlines. It's a tough game to call.

6. Business Insights & Recommendations

1. Strong Long-Term Growth in Technology Stocks:

The stock price trend analysis shows that all four technology companies experienced significant long-term growth between 2019 and 2026 despite temporary market corrections. Microsoft (MSFT) maintained the highest stock price throughout most of the analysis period, reflecting its consistent market performance.

2. High Trading Activity in Apple:

Apple (AAPL) recorded the highest average trading volume among the selected companies. Higher trading volume indicates greater market liquidity and investor participation, making Apple one of the most actively traded stocks in the dataset.

3. Daily Returns are Highly Unpredictable:

The distribution of daily returns is concentrated around zero with occasional extreme positive and negative values. This suggests that short-term stock movements are highly volatile and difficult to predict accurately using historical price information alone.

4. Market Risk Varies Over Time:

The 7-day rolling volatility analysis identified several periods of increased market uncertainty where price fluctuations became significantly larger. Monitoring volatility can help investors understand changing market risk and adjust investment strategies accordingly.

5. Technical indicators improve market understanding

Feature engineering using Daily Return, Moving Averages (MA-7 and MA-30), RSI, Rolling Volatility, and Lag Features provided additional information beyond raw stock prices. These indicators capture market momentum, trend direction, and risk, making them valuable for predictive analytics.

6. Machine Learning Supports Financial Decision-Making

Among the evaluated regression models, Random Forest achieved the lowest prediction error (RMSE = 1.986, MAE = 1.387) and was selected as the final model. Although predicting short-term market returns remains challenging due to market uncertainty, the model demonstrates how machine learning can assist investors by identifying patterns in historical market behaviour.

7. Future Scope:

Future improvements could include integrating financial news sentiment, macroeconomic indicators, company fundamentals, and deep learning models such as LSTM to further enhance predictive performance and support more comprehensive investment analysis.

7. Deliverables Checklist

- ☒ Raw Dataset
- ☒ Cleaned Dataset
- ☒ ETL Pipeline
- ☒ SQLite Database
- ☒ Exploratory Data Analysis (EDA)

- ☒ Feature Engineering
- ☒ Machine Learning Models
- ☒ Model Evaluation
- ☒ Interactive Streamlit Dashboard
- ☒ Power BI Dashboard
- ☒ Business Insights Report